THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** |
Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 90°	Reason with the angle relations.
2	**(b)** 108	Find the first multiple of 9.
3	**(b)** 3	List the triples $a \le b \le c$ with $a + b > c$.
4	**(c)** $\frac{1}{4}$	Work the bracket first.
5	**(a)** 120°	Share 180° in the given ratio.
6	**(c)** 9873	Start from 9876 and test downwards.
7	**(a)** 40°	Base angle $= 180^\circ - e$; apex $= 180^\circ - 2\times$ base.
8	**(c)** 135	Find the number first.
9	**(d)** 110°	Complement $= 90^\circ - a$; supplement of that $= 180^\circ - (90^\circ - a)$.
10	**(c)** 5	Repeat the routine and count.
11	**(a)** 90°	Reason with the angle sum and exterior angle property.
12	**(c)** 30	Time for the whole = time for the part ÷ fraction.
13	**(a)** Both A and R are true and R is the correct explanation of A.	90 - 35 = 55.
14	**(a)** Both A and R are true and R is the correct explanation of A.	2m + 1 + 2n + 1 = 2(m + n + 1), which is even.
15	**(a)** Both A and R are true and R is the correct explanation of A.	3 + 4 = 7 < 8, so the rule fails.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Let the angle be y: $180 - y = 3(90 - y)$; $2y = 90$ (1); $y = 45^\circ$ (1).	1+1
17	$11(a + b) = 110$ so $a + b = 10$; $9(a - b) = 36$ so $a - b = 4$ (1); $a = 7$, $b = 3$; Number $= 73$ (1).	1+1
18	Check where the equal parts lie in each triangle; No: SSA: the equal angle is not between the equal sides, so this is not a congruence rule (1); A different triangle with the same given parts can be drawn (1).	1+1
19	Sum $= 1\frac{1}{6}$; Difference $= \frac{1}{2}$ (1); Quotient $= 2\frac{1}{3}$; Final $= 1\frac{1}{6}$ (1).	1+1
20	Let the complement be y; the angle is $3y + 6$ and $y + (3y + 6) = 90$; $4y = 84$, so $y = 21$ (1); The angle is 69° (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (35 + 45 + 50)^\circ = 50^\circ$ (1); Kinds: 35° acute, 45° acute, 50° acute, 50° acute (1).	1+1+1
22	Digit sum $= 13 + 2 \times *$; Need a multiple of 9 (1); Possible $*$: 7 (1); Numbers: 47376 (1).	1+1+1
23	Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $4x + 60 + 3x + 100 = 230$ (1); $x = 10$; Exterior angles 100° and 130° (1); $\angle B = 80^\circ$, $\angle C = 50^\circ$ (1).	1+1+1
24	Ravi's one day work $= \frac{1}{6}$, Sita's $= \frac{1}{3}$; Together in one day $= \frac{1}{2}$ (1); Days $= 1 \div \frac{1}{2}$ (1); $= 2$ days (1).	1+1+1
	OR: One part $= 180^\circ \div 5 = 36^\circ$ (1); Smaller $= 36^\circ$ and larger $= 144^\circ$ (1); Complement of the smaller $= 54^\circ$ and supplement of the larger $= 36^\circ$ (1).	1+1+1
25	$5321 - 1235 = 4086$; $8640 - 0468 = 8172$; $8721 - 1278 = 7443$; $7443 - 3447 = 3996$; $9963 - 3699 = 6264$; $6642 - 2466 = 4176$; $7641 - 1467 = 6174$ (1); Steps $= 7$ and the sequence is $1235, 4086, 8172, 7443, 3996, 6264, 4176, 6174$ (2).	1+2
	OR: Base b, equal sides $b + 2$; $b + 2(b + 2) = 40$, so $3b = 36$ and $b = 12$ (1); Sides $14, 14, 12$ cm (1); $12 + 14 = 26 > 14$, so the triangle exists (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) 65° (1). (ii) 65° (1). (iii) $180^\circ - 65^\circ = 115^\circ$ (1). (iv) Four angles are 65° (1) and four angles are 115° (1).	1+1+1+2
27	Exterior angles add up to 360°; one part $= 360^\circ \div 12 = 30^\circ$ (1); Exterior angles 90°, 120°, 150° (1); Interior angles 90°, 60°, 30° (1); Check: 180° (1); The triangle is right-angled (1).	1+1+1+1+1
	OR: Sold $= 60$ acres; remainder $= 180$ (1); Given to son $= 60$ acres; remainder $= 120$ (1); Sugarcane $= \frac{3}{4} \times 120 = 90$ acres (1); Unplanted $= 120 - 90 = 30$ acres (1); As a fraction of the whole land, unplanted $= \frac{1}{8}$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{1}{3}$	Unpainted = (unpainted before) $\times$ (what is not painted of it).
2	**(d)** 55°	Larger + smaller $= 90^\circ$ and larger − smaller $= d$.
3	**(a)** even	Test with small values.
4	**(c)** no triangle can be formed	Check the sum of the two shorter sides.
5	**(b)** greater than 1	Reason about the size of products and quotients.
6	**(d)** 80°	Larger + smaller $= 180^\circ$.
7	**(a)** 21	Triangular numbers: $n(n+1)/2$.
8	**(d)** 10	Base $= P - 2s$.
9	**(b)** 2	Multiply the fraction by the mixed number.
10	**(d)** 120°	Total $= 180^\circ$ in 3 equal parts.
11	**(a)** 3	Add the reverse each time.
12	**(d)** 30°	They add up to 90°.
13	**(a)** Both A and R are true and R is the correct explanation of A.	6/12 = 1/2.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but equality of vertically opposite angles does not explain the linear pair.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	5 + 8 + 3 + 2 = 18 is divisible by 9, so A is true. R is true but does not explain A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	If n is even, $n(n+1)$ is even $\times$ odd, so even; If n is odd, $n+1$ is even, so the product is even (1); For $n = 13$: $13 \times 14 = 182$, which is even (1).	1+1
17	$\angle ACB = 180^\circ - 115^\circ = 65^\circ$; $\angle B = \angle ACB = 65^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2 \times 65^\circ = 50^\circ$ (1).	1+1
18	Left $= 1 - \frac{2}{7} = \frac{5}{7}$ of the number; Number $= 35 \div \frac{5}{7} = 49$ (1); Removed $= 14$ (1).	1+1
19	Larger $= (180 + 18) \div 2 = 99^\circ$; Smaller $= (180 - 18) \div 2 = 81^\circ$ (1); Complement of the smaller $= 90^\circ - 81^\circ = 9^\circ$ (1).	1+1
20	Digit sum $= 15$; Divisibility by 3 is satisfied by every arrangement (1); Largest arrangement: 7431 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	First distance $= 9$ km; Second distance $= 15$ km (1); Total $= 24$ km in 6 hours (1); Average speed $= 4$ km/h (1).	1+1+1
22	Other angle $= 180^\circ - 40^\circ = 140^\circ$ (1); Halves: 20° and 70° (1); Angle between the bisectors $= 20^\circ + 70^\circ = 90^\circ$ (1).	1+1+1
23	Try 2, 3, 4, ... numbers in turn (1); $7 + 8$; $4 + 5 + 6$; $1 + 2 + 3 + 4 + 5$ (1); Number of ways $= 3$ (1).	1+1+1
24	Third side between 13 and 29; Multiples of 5 in this range: $15, 20, 25$ (1); Largest third side 25 cm (1); Largest perimeter $= 8 + 21 + 25 = 54$ cm (1).	1+1+1
	OR: Pieces $= 20 \div \frac{5}{3} = 12$; Full pieces $= 12$ (1); Used $= 20$ m; Left $= 0$ m (1); Final statement (1).	1+1+1
25	Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (20 + 75 + 30)^\circ = 55^\circ$ (1); Kinds: 20° acute, 75° acute, 30° acute, 55° acute (1).	1+1+1
	OR: Digit sum $= 10 + 2 \times *$; Need a multiple of 9 (1); Possible $*$: 4 (1); Numbers: 74241 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $4 + 5 + 6$ (1). (ii) $6 + 7 + 8 = 21$ (1). (iii) Yes; the sum is $(n-1) + n + (n+1) = 3n$ (1). (iv) $10 + 11 + 12 = 33$ (1) and $16 + 17 = 33$ (1).	1+1+1+2

27	Corresponding parts of congruent triangles are equal (1); $3x + 10 = 5x - 20$, so $2x = 30$ and $x = 15$ (1); $\angle A = 55^\circ$ (1); $2y + 4 = 3y - 6$, so $y = 10$ and $AB = 24$ cm (1); $\angle C = 180^\circ - 55^\circ - 40^\circ = 85^\circ = \angle F$ (1).	1+1+1+1+1
	OR: A: $\frac{1}{6}$, B: $\frac{1}{12}$ per hour (1); Drain: $\frac{1}{24}$ per hour (emptying) (1); Net per hour $= \frac{5}{24}$ (1); Time $= 1 \div \frac{5}{24} = 4\frac{4}{5}$ hours (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	List a, a, b with $b = P - 2a$ and $b < 2a$.
2	**(d)** $\frac{1}{3}$	Work the bracket first.
3	**(d)** 108°	Share 180° in the given ratio.
4	**(d)** 1008	Find the first multiple of 9.
5	**(c)** 3	List the triples $a \le b \le c$ with $a + b > c$.
6	**(c)** 84	Find the number first.
7	**(b)** 125°	Complement $= 90^\circ - a$; supplement of that $= 180^\circ - (90^\circ - a)$.
8	**(d)** 9875	Start from 9876 and test downwards.
9	**(a)** 70°	Base angle $= 180^\circ - e$; apex $= 180^\circ - 2\times$ base.
10	**(d)** 20	Time for the whole = time for the part ÷ fraction.
11	**(c)** 225°	A complete turn is 360°.
12	**(b)** 7	Repeat the routine and count.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Two right angles would make 180 degrees with nothing left for the third angle. Both true, but R does not explain A.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	(3/4) x 2 = 3/2 and (1/2)(3/4) = 3/8. Both true but separate facts.
15	**(c)** A is true but R is false.	A is true. Vertically opposite angles are equal, they do not always add up to 180 degrees, so R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$b = P - 2a = 20 - 2a$ and $b < 2a$ so $a > 5$; Also $b > 0$ so $a < 10$ (1); $6, 6, 8$; $7, 7, 6$; $8, 8, 4$; $9, 9, 2$; Number of triangles $= 4$ (1).	1+1
17	Mean $= \frac{1}{2}\left(\frac{1}{3} + \frac{1}{2}\right)$; $= \frac{5}{12}$ (1); Distance from $\frac{1}{3}$: $\frac{1}{12}$; Distance from $\frac{1}{2}$: $\frac{1}{12}$ (equal) (1).	1+1
18	Let the angle be y: $180 - y = 4(90 - y)$; $3y = 180$ (1); $y = 60^\circ$ (1).	1+1
19	$11(a + b) = 143$ so $a + b = 13$; $9(a - b) = 45$ so $a - b = 5$ (1); $a = 9$, $b = 4$; Number $= 94$ (1).	1+1
20	Check where the equal parts lie in each triangle; No: AAA gives the same shape but not necessarily the same size (1); A different triangle with the same given parts can be drawn (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $5x + 50 + 4x + 62 = 220$ (1); $x = 12$; Exterior angles 110° and 110° (1); $\angle B = 70^\circ$, $\angle C = 70^\circ$ (1).	1+1+1
22	Ravi's one day work $= \frac{1}{10}$, Sita's $= \frac{1}{15}$; Together in one day $= \frac{1}{6}$ (1); Days $= 1 \div \frac{1}{6}$ (1); $= 6$ days (1).	1+1+1
23	One part $= 180^\circ \div 9 = 20^\circ$ (1); Smaller $= 20^\circ$ and larger $= 160^\circ$ (1); Complement of the smaller $= 70^\circ$ and supplement of the larger $= 20^\circ$ (1).	1+1+1
24	$9521 - 1259 = 8262$; $8622 - 2268 = 6354$; $6543 - 3456 = 3087$; $8730 - 0378 = 8352$; $8532 - 2358 = 6174$ (1); Steps $= 5$ and the sequence is $1259, 8262, 6354, 3087, 8352, 6174$ (2).	1+2
	OR: Base b, equal sides $b + 3$; $b + 2(b + 3) = 45$, so $3b = 39$ and $b = 13$ (1); Sides $16, 16, 13$ cm (1); $13 + 16 = 29 > 16$, so the triangle exists (1).	1+1+1
25	First distance $= 42$ km; Second distance $= 17\frac{1}{2}$ km (1); Total $= 59\frac{1}{2}$ km in 6 hours (1); Average speed $= 9\frac{11}{12}$ km/h (1).	1+1+1
	OR: Other angle $= 180^\circ - 70^\circ = 110^\circ$ (1); Halves: 35° and 55° (1); Angle between the bisectors $= 35^\circ + 55^\circ = 90^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) No, as $4 + 5 = 9$ is not greater than 9 (1). (ii) Yes, as the sum of any two sides exceeds the third (1). (iii) Yes, $7 + 7 = 14 > 12$; it is isosceles (1). (iv) The third side is greater than $9 - 6 = 3$ cm and less than $9 + 6 = 15$ cm (1); whole numbers 4 to 14, i.e. 11 lengths (1).	1+1+1+2
27	Exterior angles add up to 360°; one part $= 360^\circ \div 15 = 24^\circ$ (1); Exterior angles 96°, 120°, 144° (1); Interior angles 84°, 60°, 36° (1); Check: 180° (1); The triangle is acute-angled (1).	1+1+1+1+1
	OR: Sold $= 144$ acres; remainder $= 216$ (1); Given to son $= 36$ acres; remainder $= 180$ (1); Sugarcane $= \frac{4}{5} \times 180 = 144$ acres (1); Unplanted $= 180 - 144 = 36$ acres (1); As a fraction of the whole land, unplanted $= \frac{1}{10}$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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